Anesthesia for Central Nervous System Diseases V2 – MCQ

Source: Short Textbook of Anesthesia – Ajay Yadav

Anesthesia for Central Nervous System Diseases

50 Questions • 30 Minutes

Instructions:
• Each question has 4 options (A, B, C, D).
• The correct answer is shown on the right side of each question.
• Detailed explanations with textbook page references are at the end.

Questions

Q1. Parkinson's disease occurs due to progressive loss of which neurotransmitter in the brain? **A.** Serotonin **B.** Dopamine **C.** Acetylcholine **D.** GABA	**Ans: B**
Q2. Why are Parkinson's disease patients prone to arrhythmias perioperatively? **A.** Due to electrolyte imbalance **B.** Due to increased levels of dopamine in body (levodopa gets converted to dopamine) **C.** Due to vagal stimulation **D.** Due to coronary artery disease	**Ans: B**
Q3. Parkinson's disease patients develop hypotension due to: **A.** Blood loss **B.** Depletion of catecholamines by chronically elevated dopamine levels **C.** Adrenal insufficiency **D.** Cardiac failure	**Ans: B**
Q4. Parkinson's disease patients are volume depleted because of: **A.** Poor oral intake only **B.** Dopaminergic effect on kidneys **C.** Excessive sweating **D.** Diarrhea from medications	**Ans: B**
Q5. Which anesthetic drugs should be avoided in Parkinson's disease as they increase muscle tone? **A.** Propofol and thiopentone **B.** Ketamine and alfentanil **C.** Midazolam and fentanyl **D.** Isoflurane and sevoflurane	**Ans: B**
Q6. Regarding Alzheimer's disease and general anesthesia, the textbook states: **A.** General anesthesia definitely worsens dementia **B.** The rising concern that general anesthesia worsens dementia is not yet fully substantiated by studies **C.** General anesthesia improves cognitive function **D.** Only regional anesthesia must be used	**Ans: B**

Q7. The major concern of Alzheimer's disease patients presenting for surgery includes all EXCEPT: **A.** Geriatric and dementic **B.** Non-cooperative **C.** Prone to postoperative psychosis **D.** Prone to malignant hyperthermia	**Ans: D**

Q8. In epileptic patients, anticonvulsants especially affect which organ system requiring preoperative assessment? **A.** Renal function **B.** Liver function **C.** Cardiac function **D.** Pulmonary function	**Ans: B**

Q9. Which of the following drugs has epileptogenic potential due to its preservative? **A.** Propofol **B.** Ketamine **C.** Etomidate **D.** Thiopentone	**Ans: B**

Q10. Atracurium should be used with caution in epileptics because of: **A.** Histamine release **B.** Its metabolite Laudanosine has epileptogenic potential **C.** Prolonged neuromuscular block **D.** Direct CNS stimulation	**Ans: B**

Q11. Which induction agent is the agent of CHOICE for epileptic patients? **A.** Methohexitone **B.** Ketamine **C.** Thiopentone **D.** Etomidate	**Ans: C**

Q12. Phenytoin and carbamazepine can potentiate the effect of: **A.** Depolarizing muscle relaxants **B.** Non-depolarizing muscle relaxants **C.** Opioids **D.** Benzodiazepines	**Ans: B**

Q13. Spinal/epidural anesthesia in epileptic patients can be given safely if: **A.** Patient is on multiple anticonvulsants **B.** There are no signs and symptoms of raised intracranial tension (ICT) **C.** Patient had seizure more than 24 hours ago **D.** Patient is on phenytoin only	**Ans: B**

Q14. After a cerebrovascular accident, elective surgery should be deferred for: **A.** 3 months **B.** 6 months **C.** 9 months **D.** 12 months	**Ans: C**

Q15. If elective surgery is performed within 3 months after stroke, the risk of complications is how many times higher? **A.** 3 times **B.** 5 times **C.** 9 times **D.** 14 times	**Ans: D**

Q16. If elective surgery is performed within 6 months after stroke, the risk is how many times higher? **A.** 2 times **B.** 5 times **C.** 10 times **D.** 14 times	**Ans: B**

Q17. Suxamethonium is contraindicated in stroke patients because of: **A.** Risk of malignant hyperthermia **B.** Extrajunctional receptor proliferation causing hyperkalemia **C.** Histamine release **D.** Prolonged action	**Ans: B**

Q18. Patients with history of transient ischemic attacks must have which investigation before considering anesthesia? **A.** CT brain **B.** MRI brain **C.** Carotid Doppler **D.** EEG	**Ans: C**

Q19. In patients with chronic headache, analgesics especially affect which organ system? **A.** Liver function **B.** Renal function **C.** Cardiac function **D.** Hematologic function	**Ans: B**

Q20. Chronic headache patients are prone to develop post-spinal headache, therefore: **A.** Use the smallest needle **B.** Spinal should be avoided **C.** Prophylactic blood patch should be given **D.** Use hyperbaric solution	**Ans: B**

Q21. In Multiple Sclerosis, which type of regional anesthesia can precipitate the disease? **A.** Epidural anesthesia **B.** Spinal anesthesia **C.** Peripheral nerve blocks **D.** Brachial plexus block	**Ans: B**
Q22. In Multiple Sclerosis, peripheral nerve blocks: **A.** Are absolutely contraindicated **B.** Can be given safely as peripheral nerves are not involved in MS **C.** Should only be given under ultrasound guidance **D.** Require reduced local anesthetic doses	**Ans: B**
Q23. In Multiple Sclerosis patients, immunosuppressant drugs especially affect which evaluation? **A.** Liver function **B.** Hematologic evaluation **C.** Cardiac evaluation **D.** Pulmonary function	**Ans: B**
Q24. During general anesthesia in MS patients, which aspect is most important to prevent disease precipitation? **A.** Avoiding all opioids **B.** Blunting stress response to intubation and maintaining good depth of anesthesia **C.** Using total intravenous anesthesia only **D.** Keeping the patient hypothermic	**Ans: B**
Q25. Hyperthermia in Multiple Sclerosis patients: **A.** Has no effect on the disease **B.** Is a very important risk factor to precipitate MS and must be avoided at any cost **C.** Is beneficial for recovery **D.** Only affects patients in relapse	**Ans: B**
Q26. In syringomyelia, suxamethonium must not be used because: **A.** Risk of malignant hyperthermia **B.** There is lower motor neuron (LMN) paralysis **C.** Histamine release **D.** Pseudocholinesterase deficiency	**Ans: B**
Q27. In syringomyelia, response to non-depolarizing muscle relaxants is: **A.** Reduced (resistance) **B.** Normal **C.** Exaggerated **D.** Unpredictable but not exaggerated	**Ans: C**

Q28. Central neuraxial blocks in syringomyelia should be avoided because of: **A.** Coagulopathy **B.** Neurological deficits and possibility of raised ICT **C.** Technical difficulty only **D.** Risk of infection	**Ans: B**
Q29. Associated scoliosis in syringomyelia can cause which problem during general anesthesia? **A.** Difficult IV access **B.** Ventilation perfusion mismatch **C.** Cardiac tamponade **D.** Pneumothorax	**Ans: B**
Q30. In ALS (Motor Neuron Disease), central neuraxial anesthesia: **A.** Is the preferred technique **B.** Can exacerbate the disease **C.** Has no effect on the disease **D.** Is mandatory for lower limb surgery	**Ans: B**
Q31. In Guillain-Barré Syndrome, central neuraxial blocks can worsen the disease due to: **A.** Coagulopathy **B.** Double crush syndrome **C.** Autonomic dysfunction **D.** Hypotension	**Ans: B**
Q32. In autonomic dysfunction, hypotension must be treated by: **A.** Ephedrine (indirect acting vasopressor) **B.** Phenylephrine (directly acting vasopressor) **C.** Fluid bolus only **D.** Atropine	**Ans: B**
Q33. Why is ephedrine NOT the vasopressor of choice in autonomic dysfunction? **A.** It causes bradycardia **B.** It acts by releasing catecholamines which are already depleted **C.** It causes bronchospasm **D.** It has a very short duration	**Ans: B**
Q34. In acute spinal cord transection, the spinal shock phase lasts for: **A.** 24-48 hours **B.** 1-3 weeks **C.** 3-6 months **D.** 6-12 months	**Ans: B**

Q35. Patients with chronic spinal cord transection above T6 have no sensations below transection. Why do they still need anesthesia for surgery? **A.** For anxiolysis only **B.** Surgery is a potent stimulus for developing autonomic hyper-reflexia **C.** To prevent awareness **D.** For muscle relaxation only	**Ans: B**

Q36. Which anesthetic techniques can prevent autonomic hyper-reflexia in chronic spinal cord transection? **A.** Only general anesthesia **B.** Regional anesthesia and deep general anesthesia **C.** Only local infiltration **D.** Monitored anesthesia care	**Ans: B**

Q37. Regarding MAO inhibitors in psychiatric patients, the CURRENT recommendation is: **A.** Stop 2-3 weeks before surgery **B.** All antidepressants including MAO inhibitors should be continued **C.** Stop only MAO inhibitors, continue others **D.** Switch to SSRIs 4 weeks before surgery	**Ans: B**

Q38. Antidepressants increase the level of brain neurotransmitters, which causes: **A.** Decreased anesthetic requirement **B.** Increased anesthetic requirement **C.** No change in anesthetic requirement **D.** Resistance to muscle relaxants	**Ans: B**

Q39. Acute intake of tricyclic antidepressants (up to 3 weeks) increases which catecholamines, leading to arrhythmias? **A.** Dopamine and serotonin **B.** Epinephrine and norepinephrine **C.** Acetylcholine and GABA **D.** Histamine and bradykinin	**Ans: B**

Q40. In patients on acute TCAs, which drugs should NOT be used? **A.** Propofol, fentanyl, atracurium **B.** Pancuronium, ketamine, local anesthetics with adrenaline **C.** Thiopentone, morphine, vecuronium **D.** Sevoflurane, remifentanil, cisatracurium	**Ans: B**

Q41. Chronic use of antidepressants causes catecholamine depletion, which increases the risk of: **A.** Hypertension **B.** Hypotension **C.** Tachycardia **D.** Bronchospasm	**Ans: B**

Q42. Patients on MAO inhibitors must NOT receive which opioid? **A.** Morphine **B.** Fentanyl **C.** Pethidine (Meperidine) **D.** Remifentanil	**Ans: C**
Q43. The induction agent of choice for Electroconvulsive Therapy (ECT) is: **A.** Thiopentone **B.** Propofol **C.** Methohexital **D.** Ketamine	**Ans: C**
Q44. During ECT, when succinylcholine is used, a tourniquet is applied on one hand before giving succinylcholine in order to: **A.** Prevent IV extravasation **B.** See the seizure activity in that hand **C.** Reduce the dose required **D.** Prevent hyperkalemia	**Ans: B**
Q45. Lithium, used for mania, can produce which endocrine side effect? **A.** Cushing's syndrome **B.** Diabetes insipidus and electrolyte imbalance **C.** Hypothyroidism only **D.** Addison's disease	**Ans: B**
Q46. Regarding lithium and surgery, the CURRENT recommendation is: **A.** Stop 48 hours before surgery **B.** Stop 1 week before surgery **C.** Continue lithium **D.** Replace with valproate	**Ans: C**
Q47. In schizophrenic patients, antipsychotic drugs make them more vulnerable for which complication during anesthesia? **A.** Malignant hyperthermia **B.** Hypotension (alpha blocking property) and QTc prolongation **C.** Bronchospasm **D.** Hyperkalemia	**Ans: B**
Q48. Ketamine is contraindicated in schizophrenic patients because: **A.** It causes seizures **B.** These patients can become very violent after ketamine in the postoperative period **C.** It causes hypotension **D.** It prolongs QTc interval	**Ans: B**

Q49. In drug abusers, regarding acute vs chronic drug intake and anesthetic requirement: **A.** Both increase anesthetic requirement **B.** Acute intake decreases while chronic intake increases the anesthetic requirement **C.** Both decrease anesthetic requirement **D.** Acute intake increases while chronic intake decreases	**Ans: B**

Q50. Alcoholic patients presenting for surgery must be evaluated for all EXCEPT: **A.** Dilated cardiomyopathy and cirrhosis **B.** Pancreatitis and metabolic disorders (hypoglycemia) **C.** Wernicke-Korsakoff syndrome **D.** Rheumatoid arthritis	**Ans: D**

Answer Key & Explanations

Q1. Answer: B — Dopamine

Correct Answer: B. Dopamine — Parkinson occurs due to progressive loss of dopamine deficiency in the brain. All medications including levodopa should be continued perioperatively.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

Q2. Answer: B — Due to increased levels of dopamine in body (levodopa gets converted to dopamine)

Correct Answer: B. Due to increased levels of dopamine in body (levodopa gets converted to dopamine) — PD patients are prone for arrhythmias due to increased levels of dopamine in the body as levodopa gets converted to dopamine.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

Q3. Answer: B — Depletion of catecholamines by chronically elevated dopamine levels

Correct Answer: B. Depletion of catecholamines by chronically elevated dopamine levels — Hypotension occurs due to depletion of catecholamines by chronically elevated dopamine levels. These patients are also volume depleted due to dopaminergic effect on kidneys.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

Q4. Answer: B — Dopaminergic effect on kidneys

Correct Answer: B. Dopaminergic effect on kidneys — PD patients are volume depleted due to dopaminergic effect on kidneys, therefore adequate fluid replacement is a must.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

Q5. Answer: B — Ketamine and alfentanil

Correct Answer: B. Ketamine and alfentanil — Anesthetic drugs which increase the muscle tone (ketamine, alfentanil) should be avoided in PD patients as rigidity is already a major concern.

Topic: Parkinson's Disease | Ref: Ch 20, Page 196, Topic: Parkinson's Disease

Q6. Answer: B — The rising concern that general anesthesia worsens dementia is not yet fully substantiated by studies

Correct Answer: B. The rising concern that general anesthesia worsens dementia is not yet fully substantiated by studies — This is directly stated in the textbook regarding the relationship between GA and Alzheimer's disease.

Topic: Alzheimer's Disease | Ref: Ch 20, Page 196, Topic: Alzheimer's Disease

Q7. Answer: D — Prone to malignant hyperthermia

Correct Answer: D. Prone to malignant hyperthermia — Alzheimer's patients are geriatric, dementic, non-cooperative, and prone to develop postoperative psychosis. Malignant hyperthermia is not a specific concern.

Topic: Alzheimer's Disease | Ref: Ch 20, Page 196, Topic: Alzheimer's Disease

Q8. Answer: B — Liver function

Correct Answer: B. Liver function — As anticonvulsants affect the organ systems (especially liver function), evaluation of organ function is must in preoperative assessment.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

Q9. Answer: B — Ketamine

Correct Answer: B. Ketamine — Ketamine has epileptogenic potential due to its preservative. Other drugs with epileptogenic potential include methohexitone, propofol (rare opisthotonus), etomidate (myoclonic activity), enflurane, sevoflurane.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

Q10. Answer: B — Its metabolite Laudanosine has epileptogenic potential

Correct Answer: B. Its metabolite Laudanosine has epileptogenic potential — Atracurium's metabolite Laudanosine has epileptogenic potential and should be used with caution in epileptic patients.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

Q11. Answer: C — Thiopentone

Correct Answer: C. Thiopentone — Due to its anticonvulsant property, thiopentone is the induction agent of choice for epileptics.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

Q12. Answer: B — Non-depolarizing muscle relaxants

Correct Answer: B. Non-depolarizing muscle relaxants — Phenytoin and carbamazepine can potentiate the effect of non-depolarizing muscle relaxants.

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

Q13. Answer: B — There are no signs and symptoms of raised intracranial tension (ICT)

Correct Answer: B. There are no signs and symptoms of raised intracranial tension (ICT) — Spinal/epidural can be given safely if there are no signs and symptoms of raised intracranial tension (ICT).

Topic: Epilepsy | Ref: Ch 20, Page 196, Topic: Epilepsy

Q14. Answer: C — 9 months

Correct Answer: C. 9 months — After cerebrovascular accident, elective surgery should be deferred for 9 months. The odds are stabilized at 3 times after 9 months.

Topic: Stroke | Ref: Ch 20, Page 196, Topic: Stroke

Q15. Answer: D — 14 times

Correct Answer: D. 14 times — The possibility of serious complications like another stroke, heart attack or even death is 14 times higher if elective surgery is performed within three months after stroke.

Topic: Stroke | Ref: Ch 20, Page 196, Topic: Stroke

Q16. Answer: B — 5 times

Correct Answer: B. 5 times — Risk is 5 times higher if surgery is performed within 6 months after stroke, and 14 times higher within 3 months.

Topic: Stroke | Ref: Ch 20, Page 196, Topic: Stroke

Q17. Answer: B — Extrajunctional receptor proliferation causing hyperkalemia

Correct Answer: B. Extrajunctional receptor proliferation causing hyperkalemia — Due to extrajunctional receptors, suxamethonium is contraindicated for stroke patients.

Topic: Stroke | Ref: Ch 20, Page 196, Topic: Stroke

Q18. Answer: C — Carotid Doppler

Correct Answer: C. Carotid Doppler — Patients with history of transient ischemic attacks must have at least carotid Doppler before considering for anesthesia.

Topic: Stroke | Ref: Ch 20, Page 196, Topic: Stroke

Q19. Answer: B — Renal function

Correct Answer: B. Renal function — As analgesics affect the organ systems (especially renal function), evaluation of organ function is must in preoperative assessment of chronic headache patients.

Topic: Headache | Ref: Ch 20, Page 196, Topic: Headache

Q20. Answer: B — Spinal should be avoided

Correct Answer: B. Spinal should be avoided — These patients are prone to develop post-spinal headache therefore, spinal should be avoided.

Topic: Headache | Ref: Ch 20, Page 196, Topic: Headache

Q21. Answer: B — Spinal anesthesia

Correct Answer: B. Spinal anesthesia — Spinal anesthesia (but not epidural) and stress can precipitate multiple sclerosis, probably because of direct exposure of local anesthetic to sheath of spinal cord.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

Q22. Answer: B — Can be given safely as peripheral nerves are not involved in MS

Correct Answer: B. Can be given safely as peripheral nerves are not involved in MS — As peripheral nerves are not involved in MS, peripheral nerve blocks can be given safely.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

Q23. Answer: B — Hematologic evaluation

Correct Answer: B. Hematologic evaluation — As immunosuppressant drugs affect the organ systems (especially hematologic), evaluation of organ function is must in preoperative assessment.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

Q24. Answer: B — Blunting stress response to intubation and maintaining good depth of anesthesia

Correct Answer: B. Blunting stress response to intubation and maintaining good depth of anesthesia — As stress can precipitate MS, blunting stress response to intubation and maintaining good depth of anesthesia is a must.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

Q25. Answer: B — Is a very important risk factor to precipitate MS and must be avoided at any cost

Correct Answer: B. Is a very important risk factor to precipitate MS and must be avoided at any cost — Hyperthermia is one of a very important risk factor to precipitate MS therefore, must be avoided at any cost.

Topic: Multiple Sclerosis | Ref: Ch 20, Page 197, Topic: Multiple Sclerosis

Q26. Answer: B — There is lower motor neuron (LMN) paralysis

Correct Answer: B. There is lower motor neuron (LMN) paralysis — As there is lower motor neuron (LMN) paralysis in syringomyelia, suxamethonium must not be used and exaggerated response to non-depolarizer is expected.

Topic: Syringomyelia | Ref: Ch 20, Page 197, Topic: Syringomyelia

Q27. Answer: C — Exaggerated

Correct Answer: C. Exaggerated — In syringomyelia, exaggerated response to non-depolarizer is expected due to lower motor neuron involvement.

Topic: Syringomyelia | Ref: Ch 20, Page 197, Topic: Syringomyelia

Q28. Answer: B — Neurological deficits and possibility of raised ICT

Correct Answer: B. Neurological deficits and possibility of raised ICT — Neurological deficits and possibility of raised ICT warrants the avoidance of central neuraxial blocks in syringomyelia.

Topic: Syringomyelia | Ref: Ch 20, Page 197, Topic: Syringomyelia

Q29. Answer: B — Ventilation perfusion mismatch

Correct Answer: B. Ventilation perfusion mismatch — Associated scoliosis in syringomyelia can cause ventilation perfusion mismatch during general anesthesia.

Topic: Syringomyelia | Ref: Ch 20, Page 197, Topic: Syringomyelia

Q30. Answer: B — Can exacerbate the disease

Correct Answer: B. Can exacerbate the disease — In ALS, central neuraxial can exacerbate the disease and should be avoided.

Topic: Amyotrophic Lateral Sclerosis | Ref: Ch 20, Page 197, Topic: Amyotrophic Lateral Sclerosis

Q31. Answer: B — Double crush syndrome

Correct Answer: B. Double crush syndrome — Central neuraxial blocks can worsen the already diseased nerve (double crush syndrome) in GBS, therefore should be avoided.

Topic: Guillain-Barré Syndrome | Ref: Ch 20, Page 197, Topic: Guillain-Barré Syndrome (GBS)

Q32. Answer: B — Phenylephrine (directly acting vasopressor)

Correct Answer: B. Phenylephrine (directly acting vasopressor) — Hypotension must be treated by directly acting vasopressors (phenylephrine), not by indirectly acting (ephedrine) vasopressors which act by releasing catecholamines (which are already depleted).

Topic: Autonomic Dysfunction | Ref: Ch 20, Page 197, Topic: Autonomic Dysfunction

Q33. Answer: B — It acts by releasing catecholamines which are already depleted

Correct Answer: B. It acts by releasing catecholamines which are already depleted — Indirectly acting vasopressors (ephedrine) work by releasing catecholamines, which are depleted in autonomic dysfunction, making them ineffective.

Topic: Autonomic Dysfunction | Ref: Ch 20, Page 197, Topic: Autonomic Dysfunction

Q34. Answer: B — 1-3 weeks

Correct Answer: B. 1-3 weeks — During the first 1-3 weeks patients are in spinal shock due to sympathetic cut off, and are very vulnerable for hypotension and bradycardia.

Topic: Spinal Cord Transection | Ref: Ch 20, Page 197, Topic: Spinal Cord Transection

Q35. Answer: B — Surgery is a potent stimulus for developing autonomic hyper-reflexia

Correct Answer: B. Surgery is a potent stimulus for developing autonomic hyper-reflexia — Although these patients have no sensations below transection, they need anesthesia because surgery is a potent stimulus for developing autonomic hyper-reflexia.

Topic: Spinal Cord Transection | Ref: Ch 20, Page 197, Topic: Spinal Cord Transection

Q36. Answer: B — Regional anesthesia and deep general anesthesia

Correct Answer: B. Regional anesthesia and deep general anesthesia — Regional anesthesia and deep GA can prevent autonomic hyper-reflexia in patients with chronic spinal cord transection above T6.

Topic: Spinal Cord Transection | Ref: Ch 20, Page 197, Topic: Spinal Cord Transection

Q37. Answer: B — All antidepressants including MAO inhibitors should be continued

Correct Answer: B. All antidepressants including MAO inhibitors should be continued — Contrary to previous recommendation of stopping MAO inhibitors 2-3 weeks prior, all antidepressants including MAO inhibitors should be continued.

Topic: Psychiatric Disorders – Depression | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

Q38. Answer: B — Increased anesthetic requirement

Correct Answer: B. Increased anesthetic requirement — Antidepressants increase the level of brain neurotransmitters, increasing the anesthetic requirement.

Topic: Psychiatric Disorders – Depression | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

Q39. Answer: B — Epinephrine and norepinephrine

Correct Answer: B. Epinephrine and norepinephrine — Acute intake of tricyclic antidepressants (up to 3 weeks) increases the epinephrine and norepinephrine, increasing chances of arrhythmias (particularly with halothane) and hypertensive crisis.

Topic: Psychiatric Disorders – Depression | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

Q40. Answer: B — Pancuronium, ketamine, local anesthetics with adrenaline

Correct Answer: B. Pancuronium, ketamine, local anesthetics with adrenaline — Drugs like pancuronium, ketamine and local anesthetics with adrenaline should not be used due to sympathomimetic effects potentiating catecholamine excess.

Topic: Psychiatric Disorders – Depression | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

Q41. Answer: B — Hypotension

Correct Answer: B. Hypotension — Chronic use causes catecholamine depletion increasing the risk of hypotension. This is opposite to acute TCA intake which causes sympathetic excess.

Topic: Psychiatric Disorders – Depression | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

Q42. Answer: C — Pethidine (Meperidine)

Correct Answer: C. Pethidine (Meperidine) — Patients on MAO inhibitors must not receive pethidine due to the risk of fatal serotonin syndrome.

Topic: Psychiatric Disorders – Depression | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

Q43. Answer: C — Methohexital

Correct Answer: C. Methohexital — Methohexital, being epileptogenic, is the induction agent of choice for ECT. Propofol may shorten seizure duration.

Topic: Psychiatric Disorders – ECT | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

Q44. Answer: B — See the seizure activity in that hand

Correct Answer: B. See the seizure activity in that hand — If succinylcholine is used to prevent muscle contraction, tourniquet should be applied in one hand before giving succinylcholine to see the seizure activity.

Topic: Psychiatric Disorders – ECT | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

Q45. Answer: B — Diabetes insipidus and electrolyte imbalance

Correct Answer: B. Diabetes insipidus and electrolyte imbalance — Patients on lithium can develop diabetes insipidus and electrolyte imbalance. Current recommendation is to continue lithium perioperatively.

Topic: Psychiatric Disorders – Mania | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

Q46. Answer: C — Continue lithium

Correct Answer: C. Continue lithium — In contrary to older guidelines of stopping lithium 48 before surgery, current recommendation is to continue lithium. Lamotrigine is the other drug approved for bipolar disorder.

Topic: Psychiatric Disorders – Mania | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

Q47. Answer: B — Hypotension (alpha blocking property) and QTc prolongation

Correct Answer: B. Hypotension (alpha blocking property) and QTc prolongation — Antipsychotic patients are more vulnerable for hypotension (alpha blocking property), arrhythmias (QTc prolongation and torsade de pointes).

Topic: Psychiatric Disorders – Schizophrenia | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

Q48. Answer: B — These patients can become very violent after ketamine in the postoperative period

Correct Answer: B. These patients can become very violent after ketamine in the postoperative period — Ketamine is contraindicated because these patients can become very violent after ketamine in the postoperative period.

Topic: Psychiatric Disorders – Schizophrenia | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

Q49. Answer: B — Acute intake decreases while chronic intake increases the anesthetic requirement

Correct Answer: B. Acute intake decreases while chronic intake increases the anesthetic requirement — Acute drug intake decreases while chronic intake increases the anesthetic requirement due to tolerance and cross-tolerance.

Topic: Psychiatric Disorders – Drug Abuse | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders

Q50. Answer: D — Rheumatoid arthritis

Correct Answer: D. Rheumatoid arthritis — Alcoholic patients must be evaluated for dilated cardiomyopathy, cirrhosis, pancreatitis, metabolic disorders (hypoglycemia), nutritional disorders (Wernicke-Korsakoff syndrome). Rheumatoid arthritis is not associated.

Topic: Psychiatric Disorders – Drug Abuse | Ref: Ch 20, Page 197, Topic: Psychiatric Disorders